

Model Validation, Overfitting Control

California Housing Price Prediction

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Overfitting Analysis:

Default Decision Tree: High overfitting (Train RMSE low, Test RMSE high)

Train RMSE: 0.000

Test RMSE: 0.703

Overfitting Gap: -0.703 (Large gap = Overfitting)

Cross-Validation & Hyperparameter Tuning:

```
Cross Validation RMSE: 0.896  
CV Score: [0.88767617 0.82781619  
0.89845112 0.94507041 0.91950207]
```

```
Best Parameters: {'max_depth': 10, 'min_samples_leaf': 4,  
'min samples split': 2}  
Best CV Score: 0.6218739737028046
```

Model Comparison:

	RMSE	R2
Linear Regression	0.746	0.576
Ridge Regression	0.746	0.576
Tuned Decision Tree	0.639	0.688

Final Justification & Learnings:

Selected Model: Tuned Decision Tree Regressor

Why this model was selected?

- It achieved the **highest R^2 score** and lowest RMSE on the test set after hyperparameter tuning.
- It outperformed baseline Linear Regression and Ridge Regression.

How overfitting was reduced?

- Default Decision Tree showed large gap between Train and Test RMSE (clear overfitting).
- After tuning `max_depth`, `min_samples_split`, and `min_samples_leaf` using GridSearchCV, the gap reduced significantly.
- This makes the model more generalizable.

Why trust Cross-Validation results?

- Used 5-Fold Cross Validation instead of single train-test split.
- CV score is more reliable and closer to real-world performance.

Trade-offs:

- Slightly more complex than Linear Regression but much better performance.
- Still interpretable and fast for prediction.

Conclusion: The Tuned Decision Tree is the best choice for this project because it balances performance and generalization. This follows industry best practices for building reliable ML models.